SRI SIVASUBRAMANIYA NADAR COLLEGE OF ENGINEERING
(An Autonomous Institution - Affiliated to Anna University)
Rajiv Gandhi Salai, Kalavakkam - 603 110

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
B.E(CSE) PROGRAM

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

Higher education and successful career: To enable graduates to pursue higher education and research or have a successful career in industries associated with Computer Science and Engineering, or as entrepreneurs.

Competence and Life-long learning: To ensure that graduates will have the ability and attitude to acquire new skills and adapt to emerging technological changes.

Professionalism: To ensure that graduates will be professional and ethical in their work, contributing to the advancement of society.

PROGRAM OUTCOMES (POs)

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

1. To investigate challenging problems across various domains with appropriate computational techniques, construct solutions systematically and evaluate their effectiveness.
2. To apply software engineering principles and practices for building high quality software systems using contemporary computing paradigms.
3. To adopt emerging information processing technologies for producing innovative solutions to current societal problems.